Research Technical Note

Equity Futures Options

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Keywords: Futures, equity futures option, quanto option, currency translation, BAW.

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Contents

1 Introduction

This document covers the pricing analytics for Equity futures options, including the cases of currency translated options (cross-currency and quanto). The payoff of an option on a futures contract expiring at time T is given by:

$$C = N_c N \max(F_\tau - K, 0) \tag{1}$$

$$P = N_c N \max(K - F_\tau, 0), \tag{2}$$

where F_τ is the futures price underlying the option at exercise time τ ($0 \leq \tau \leq T$ for American, $\tau = T$ for European options), and K is the strike price of the option.

In the case of an ordinary option, K and F are expressed in the same currency matching the payoff currency. For a quanto option the futures contract (and the strike) is expressed in a currency different from the payoff, but it is translated into payoff currency at a fixed exchange rate $\bar{X}$. Hence:

$$C_q = N_c N/\bar{X} \max(F_\tau - K, 0) \quad (3)$$

$$P_q = N_c N/\bar{X} \max(K - F_\tau, 0), \quad (4)$$

Finally, in cross-currency options, payoff and strike are in payoff currency, while F is in underlier currency translated into payoff currency at exercise time via the prevailing exchange rate X_τ:

$$C_X = N_c N \max(F_\tau/X_\tau - K, 0) \quad (5)$$

$$P_X = N_c N \max(K - F_\tau/X_\tau, 0). \quad (6)$$

2 Definitions

	Name	Spec Variable	Required?	Default	Example
1.	Equity name		Yes	-	
2.	Underlying price	F	No	-	36.42
3.	Number of option contracts	N_c	Yes	-	100
4.	Futures expiry date	t	No	see definition	20070207
5.	Option expiry date	T	Yes	-	20080228
6.	Underlier currency		No		USD
7.	Option strike price	K	Yes	-	33.45
8.	Market price		No	-	0.76
9.	Underlying volatility	σ	No	-	19.233
10.	Volatility series		No	-	
11.	Option discount curve	r	No	Govt curve	USD Fed Funds OIS
12.	Futures financing curve	r^{fin}	No	Govt curve	USD Swap
13.	Financing Spread Curve	s^{fin}	No	0	S&P 500 Futures Financing S
14.	Equity price	S	No	-	57.60
15.	Dividend yield	q	No	0	1.2
16.	Use Historical Volatility		No	see definition	
17.	Beta	β	No	1	1.2
18.	Vol Adjustment Type		No	multiplicative	additive
19.	Number of Shares	N	No	1	100
20.	Currency translation		No		
20.1.	Payoff currency		Yes	-	EUR
20.2.	Payoff currency risk free curve	$\tilde{r}$	No	-	EUR Swap
20.3.	Underlier currency risk free curve	r	No	-	USD Swap
20.4.	Exchange rate is fixed		No		
20.5.	Fixed exchange rate	$\bar{X}$	No	1	1.2
20.6.	Exchange rate volatility	σ_X	No	-	13.2
20.7.	Exchange rate volatility series		No	-	
20.8.	Correlation	ρ	No	-	35.4

1. **Equity name** - The name of the equity.

2. **Underlying price** - The price of the equity futures contract underlying the option. If specified it is always used such that a constant spread is calibrated for forwarding Equity Price in order to match this input . If unspecified it is derived from a model as specified below.

3. **Number of option contracts** - The number of option contracts held in a position. A positive (negative)

value indicates a long (short) position.

4. **Futures expiry date** - The date at which the futures contract expires. If left unspecified, it is assumed to be equal to OptionExpiryDate.

5. **Option expiry date** - The date at which the option contract expires.

6. **Underlier currency** - The currency of the underlier. If not specified it is inferred from the Equity Name.

7. **Option strike price** - The price at which the underlying equity futures may be purchased or sold upon exercise of the option contract. Must be a positive number expressed in units of the underlier currency. If the option is currency translated but not a quanto, the strike price is expressed in units of PayoffCurrency.

8. **Market price** - The market price of the option for one unit of underlying ($N = 1$). If specified, and Volatility is not specified, an implied volatility is computed from this input. If specified together with Volatility, Volatility prevails, i.e. the price in the base scenario is computed with the user specified Volatility.

9. **Underlying volatility** - The annualized volatility used to value the option. For details on how the Volatility, VolatilitySeries, and Market Price inputs are used for pricing and calibration see the Price Calibration section below.

10. **Volatility series** - Identifier for implied volatility time-series.

11. **Option discount curve** - Discount curve used to extract risk-free rate r for option pricing. If omitted, the government curve corresponding to Underlier currency is used. Ignored if currency translation input is specified. Must be a yield curve name that matches one currently defined in the Market Data database. If the curve's status is risky in the Market Data base, or specified to be risky by the user through the use of RiskyRisklessDefn input, then the risk-free rate r will be exposed to spread risk. Negative rates can be allowed by setting allowsNegative attribute to True.

12. **Futures financing curve** - Yield curve used to extract the financing rate r^{fin} for computing the price of underlying futures contract. If omitted, the government curve corresponding to Underlier currency is used. Ignored if currency translation input is specified. Must be a yield curve name that matches one currently defined in the Market Data database. If the curve's status is risky in the Market Data base, or specified to be risky by the user through the use of RiskyRisklessDefn input, then the financing rate r^{fin} will be exposed to spread risk. Negative rates can be allowed by setting allowsNegative attribute to True.

13. **Financing Spread Curve** - The name of the financing spread curve used to extract the spread to be added to the financing rate r^{fin} from the Futures Financing Curve to obtain the 'total' financing rate to the futures' expiry date. If not specified, the financing spread curve value is set to zero.

14. **Equity price** - The price of the equity underlying the futures contract. Time series values are scaled so that the equity price used in base scenarios matches this input.

15. **Dividend yield** - Continuously-compounded dividend yield on the underlying equity, entered as an annualized percentage. If omitted it defaults to zero.

16. **Use Historical Volatility** - If selected, a volatility time series will be constructed from historical observations of equity returns. This feature exists to allow vega risk to be computed without the use of a volatility series. Each point in the constructed volatility time series is set equal to the mean zero standard deviation, estimated from equally weighted (decay factor = 1.0) daily log returns observed prior to the corresponding observation date of that volatility point. The default number of equity returns (the sampling period) used in the estimate of each volatility point is 50. This number can be modified by selecting a different sampling period via the statisticsTerm input. We allow four periods (90D, 180D, 1Y, 2Y) with the actual number of equity returns used in each case equal to 64, 128, 260, 520. If a value different from the four above is specified for statisticTerm, we use the closest term allowed.

17. **Beta** - If specified, the underlying of the option is assumed to be linked via a beta coefficient to EquityName (usually representing an index or a proxy). In this case, if the user supplies a volatility input it will be interpreted as the volatility associated to EquitName and it will be multiplied by β when used in pricing the option.

18. **Vol Adjustment Type** - It specifies if calibration to user supplied volatility or market price is done through a multiplicative or an additive factor. If omitted, it defaults to multiplicative.

19. **Number of Shares** - Number of shares underlying the futures contract. If omitted it is set to 1. The value of the option scales with NumberOfShares. Must be a positive number.

20. **Currency translation** - Fields under this section must be specified for quanto or cross currency options. An option is called a quanto if the final payoff amount is payed out in a currency different from the native currency of the underlier (the *Underlier currency*), at an FX rate fixed on the start date of the option. If the FX rate is floating the option is referred to as a cross currency option. If this tag is unspecified the option is an ordinary option without currency translation.

20.1. **Payoff currency** - The currency of the option payoff.

20.2. **Payoff currency risk free curve** - The risk free curve associated to the Payoff currency. If omitted the corresponding government curve is used.

20.3. **Underlier currency risk free curve** - The risk free curve associated to the Underlier currency. If omitted the corresponding government curve is used.

20.4. **ExchangeRateIsFixed** - Unary tag specifying that currency translation is done at a fixed exchange rate, i.e. the option is a quanto. If not specified, the FixedExchangeRate input is not used.

20.5. **Fixed exchange rate** - The predetermined FX rate used to convert the option payoff from the underlier currency to the payoff currency. Its default value is 1. Only used if ExchangeRateIsFixed is specified. It is expressed as PayoffCurrency in units of UnderlierCurrency.

20.6. **Exchange rate volatility** - Volatility of the FX rate return, expressed as an annualized percentage. If not specified, but an exchange rate volatility series is specified, the value from the volatility series is used; otherwise, the volatility is computed from historical FX returns.

20.7. **Exchange rate volatility series** - Identifier for Exchange rate volatility time-series; if not specified, the Exchange rate volatility will not be a risk factor.

20.8. **Correlation** - Correlation between the FX return (PayoffCurrency in units of UnderlierCurrency) and the return of the underlier, expressed as a percentage. If not specified, it is computed from historical time-series of the FX rate return and underlying return.

3 Analytics

For the formulas in sub-sections 3.1 and 3.2, if currency translation input is provided, the yield y, discounting rate r_T, and volatility parameters should be set to the transformed values of yield, risk free rate, and volatility respectively, defined by the parameter transformations in Sec. [3.3].

3.1 European Option on Futures

The Black-76 formula (see [2]) can be used to price European options when the underlying asset is a forward or futures contract. The underlying price, F, can be specified by the user. If left unspecified, the model price for F is obtained by:

$$F = e^{(r_t^{fin,tot} - q)t} S,$$

where S is the equity spot price, t is the time to maturity (in years) of the futures contract, and $r_t^{fin,tot}$ the corresponding 'total' financing rate that applies to the holder between analysis date and futures expiry. The total financing rate is obtained as the sum of the 'base' financing rate r^{fin} from the Futures Financing Curve and financing spread s^{fin} from the Financing Spread Curve, i.e. $r_t^{fin,tot} = r_t^{fin} + s_t^{fin}$. When Financing Spread Curve is not specified, $s^{fin} = 0$ in every market scenario.

Thus, the price of a call option is defined as

$$c = e^{-r_T T}\left(F\Phi(d_1) - K\Phi(d_2)\right) \tag{7}$$

where

$$\begin{aligned} d_1 &= \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\ d_2 &= d_1 - \sigma\sqrt{T}, \end{aligned} \tag{8}$$

with T the time to maturity of the option in years and r the corresponding risk-free interest rate. $\Phi(\cdot)$ is the cumulative univariate normal distribution

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt. \tag{9}$$

Similarly, the price for a put option is given by

$$p = e^{-r_T T}\left[K\Phi(-d_2) - F\Phi(-d_1)\right]. \tag{10}$$

3.2 American Option on Futures

In the formulas that follow, the yield y should be set equal to the rate r_T, since the underlying of the option is a futures contract, and $n = 0$.

American call options are priced using the Barone-Adesi and Whaley analytic approximation[1] as

$$C(F,T) = \begin{cases} c(F,T) + A_c\left(F/F_c^*\right)^{\gamma_c} & \text{if} \quad F < F_c^* \\ F - K & \text{if} \quad F \geq F_c^* \end{cases} \tag{11}$$

where $c(F,T)$ is the European option price (via Black-Scholes),

$$A_c = \left(F_c^*/\gamma_c\right)\left(1 - e^{-yT}\Phi[d_1(F_c^*)]\right), \tag{12}$$

$\Phi(\cdot)$ is the cumulative function given above in Eq. (9), and

$$\gamma_c = \frac{\left[-(n-1) + \sqrt{(n-1)^2 + 4m/l}\right]}{2}$$

with $m = 2r_T/\sigma^2$, $n = 2(r_T - y)/\sigma^2$, and $l = 1 - e^{-r_T T}$. The threshold value F_c^* is determined by enforcing the continuity of the approximate call value for $F = F_c^*$:

$$F_c^* - K = c(F_c^*, T) + A_c \tag{13}$$

The analogous equations for a put option are:

$$P(F,T) = \begin{cases} p(F,T) + A_p \left(F/F_p^*\right)^{\gamma_p} & \text{if} \quad F > F_p^* \\ K - F & \text{if} \quad F \leq F_p^* \end{cases} \tag{14}$$

where $p(F,T)$ is the European put option price (via Black-Scholes),

$$A_p = \left(F_p^*/\gamma_p\right)\left(e^{-yT}\Phi[-d_1(F_p^*)] - 1\right),$$

and

$$\gamma_p = \frac{\left[-(n-1) - \sqrt{(n-1)^2 + 4m/l}\right]}{2}$$

As with calls the threshold value F_p^* is determined by enforcing the continuity of the approximate put value for $F = F_p^*$:

$$K - F_p^* = p(F_p^*, T) + A_p \tag{15}$$

An iterative algorithm for determining the threshold for call (F_c^*) and puts (F_p^*) is given in [1].

3.2.1 Continuity of the Greeks in the BAW approximation

The critical boundary F_c^* in Eq. (11) is determined by the constraint that the call value $C(S,T)$ is continuous across the boundary. In addition, the coefficient A_c in this equations was defined to guarantee that the Greek Delta is continuous across the $F = F_c^*$ boundary as well. The same is true in Eq. (14) for the value of the put $P(S,T)$ at F_p^*.

To verify that Delta for the call option is continuous at $F = F_c^*$, take the first derivative of both branches of Eq. (11) with respect to F. For $F \geq F_c^*$ the derivative is one. For $F < F_c^*$ it yields

$$\begin{aligned} \frac{\partial}{\partial F}\left[c(F,T) + A_c\left(\frac{F}{F_c^*}\right)^{\gamma_c}\right] &= e^{-yT}\Phi\left(d_1(F)\right) + \gamma_c \frac{A_c}{F_c^*}\left(\frac{F}{F_c^*}\right)^{\gamma_c - 1} \\ &= e^{-yT}\Phi\left(d_1(F)\right) + \left(\frac{F}{F_c^*}\right)^{\gamma_c - 1}\left[1 - e^{-yT}\Phi\left(d_1(F_c^*)\right)\right], \end{aligned} \tag{16}$$

where we have used Eq. (12). Evaluating this expression at $F = F_c^*$ gives:

$$\begin{aligned}\frac{\partial}{\partial F}\left[c(F,T)+A_c\left(\frac{F}{F_c^*}\right)^{\gamma_c}\right]_{F=F_c^*} &= e^{-yT}\Phi\left(d_1(F_c^*)\right)+\left[1-e^{-yT}\Phi\left(d_1(F_c^*)\right)\right]\\ &= 1,\end{aligned} \tag{17}$$

showing the continuity of the Delta of the call option across the exercise boundary.

However, higher order derivatives like Gamma are not continuous across the boundary at $F = F_c^*$. This can be seen by taking the second derivative with respect to F of both branches of Eq. (11). For $F \geq F_c^*$ this is clearly zero, while for $F < F_c^*$ we have

$$\begin{aligned}\frac{\partial^2}{\partial F^2}\left[c(F,T)+A_c\left(\frac{F}{F_c^*}\right)^{\gamma_c}\right] &= \frac{\partial}{\partial F}\left[e^{-yT}\Phi\left(d_1(F)\right)+\gamma_c\frac{A_c}{F_c^*}\left(\frac{F}{F_c^*}\right)^{\gamma_c-1}\right]\\ &= \frac{\phi\left(d_1(F)\right)e^{-yT}}{S\sigma\sqrt{T}}+\gamma_c(\gamma_c-1)\frac{A_c}{(F_c^*)^2}\left(\frac{F}{F_c^*}\right)^{\gamma_c-2},\end{aligned} \tag{18}$$

where $\phi(x) = \Phi'(x)$ is the univariate normal distribution function. Evaluating this expression at $F = F_c^*$ gives

$$\begin{aligned}\frac{\partial}{\partial F}\left[c(F,T)+A_c\left(\frac{F}{F_c^*}\right)^{\gamma_c}\right]_{F=F_c^*} &= \frac{\phi\left(d_1(F_c^*)\right)e^{-yT}}{F_c^*\sigma\sqrt{T}}+\gamma_c(\gamma_c-1)\frac{A_c}{(F_c^*)^2}\\ &= \frac{\phi\left(d_1(F_c^*)\right)e^{-yT}}{F_c^*\sigma\sqrt{T}}+\frac{(\gamma_c-1)}{F_c^*}\left(1-e^{-yT}\Phi[d_1(F_c^*)]\right).\end{aligned} \tag{19}$$

This shows that, in general, the second derivatives in the two regions do not match at $F = F_c^*$ and that Gamma in the BAW pricing formula is discontinuous across the exercise boundary.

The discontinuity in Gamma is a reflection of the kink in Delta across the critical price, which is evident, for instance, in Figure 1. The kink in the Delta arises from the continually changing slope as one approaches F_c^* from the left (for a call option). For F in the vicinity of F_c^*, for example, $F \approx F_c^* - \varepsilon$ ($\varepsilon > 0$), it can be shown that starting with Eq. (16)

$$\Delta(F \approx F_c^* - \varepsilon) \approx 1 - \varepsilon g(F_c^*) + \mathcal{O}(\varepsilon^2) \tag{20}$$

for some function, $g(F_c^*)$, that can worked out explicitly. This shows that the slope of the delta for $F < F_c^*$ scales linearly with the displacement, ε, and at $F = F_c^*$ the slope becomes 1.

Is this discontinuity a manifestation of the BAW approximation, or is it an inherent property of American options in the neighborhood of the exercise boundary? To answer this question, we priced American options using a standard binomial tree algorithm (see [3]), and analysed the behavior in the neighborhood of the exercise boundary. The discontinuity in Gamma is present in the tree construction as well. Figures 1 and 2 compare the Greeks of American call options computed using the BAW formula to those computed using the binomial tree algorithm. Figure 1 confirms that Delta is continuous across the boundary at F_c^*. Figure 2 shows that Gamma is discontinuous at F_c^* for both the BAW formula and the tree algorithm. We conclude that the discontinuity is not specific to the BAW approximation and is a generic phenomenon of American-style options.

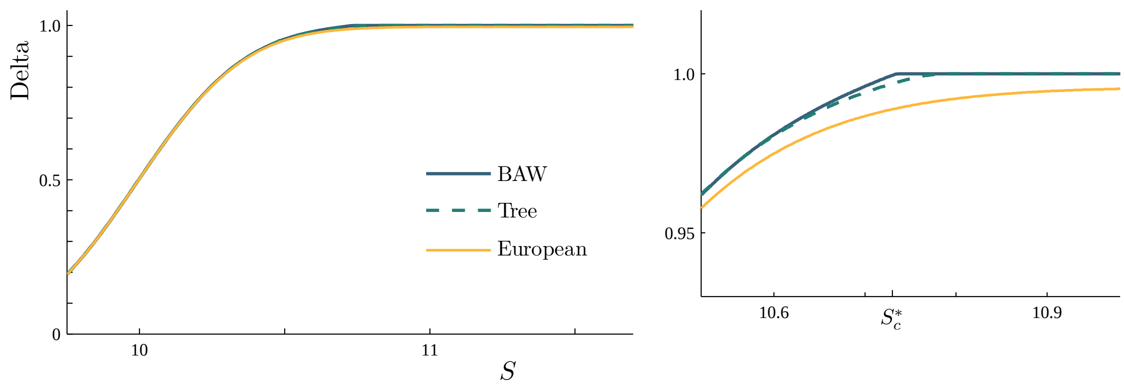

Figure 1: Delta of an American call option as a function of underlying price F for the BAW approximation and a binomial tree algorithm. The value for an equivalent European-style option is shown for reference. The right frame shows the behavior in the neighborhood of F_c^* in detail. In this example, $X = 10$, $\sigma = 10\%$, $r = y = 5\%$, $T = 1/12$, and $F_c^* \approx 10.73$.

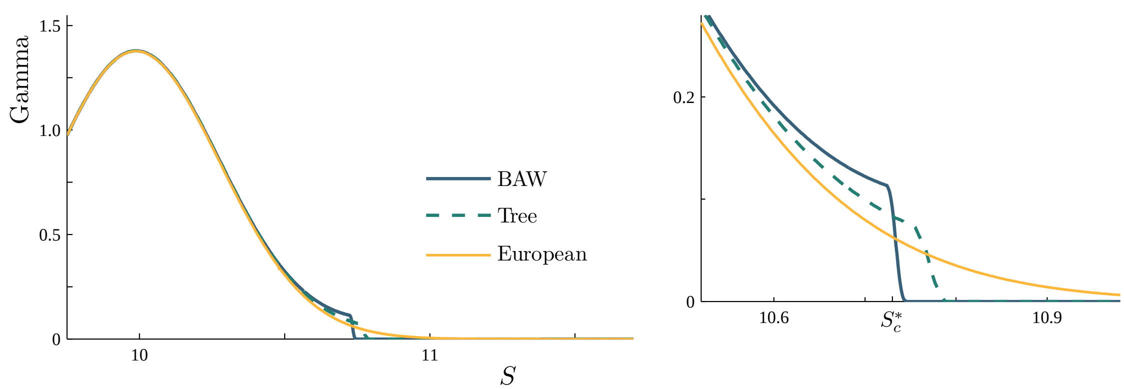

Figure 2: Gamma of an American call option as a function of underlying price F for the BAW approximation and a binomial tree algorithm. The value for an equivalent European-style option is shown for reference. The right frame shows the behavior in the neighborhood of F_c^* in detail.

One might expect that derivatives with respect to other parameters are discontinuous, too, but this is more difficult to show because F_c^* is a function of the other pricing parameters. For example, using the BAW equation for a call option, the (total) Vega of the option for $F < F_c^*$ is

$$\frac{dC(F,T)}{d\sigma} = \frac{\partial C(F,T)}{\partial \sigma} + \frac{\partial F_c^*}{\partial \sigma} \frac{\partial C(F,T)}{\partial F_c^*} \tag{21}$$

because the critical price F_c^* is a function of the volatility, as determined by solving equation 13, and $F_c^*(\sigma)$

has no simple analytic solution. However, it is easy to compute the first partial term (w.r.t. σ) and observe that it would require a non-trivial degree of cancellation with the second term to achieve $\partial C(F,T)/\partial\sigma = 0$, at $F = F_c^*$, as the second branch's solution does.

3.3 Cross currency and quanto options

Strike and payoff of cross currency options are denominated in payoff currency which is different from the underlying currency. The underlying price is converted into the payoff currency at the spot FX rate at exercise of the option, see Eq. (6).

In contrast, the strike of a quanto option is given in the underlying currency, but the payoff of the option is converted to the payoff currency at a predetermined FX rate (already known at the inception of the trade), see Eq. (4). The fixed FX rate $\bar{X}$ is often equal to 1.

The pricing formulas of the European options and the Barone-Adesi and Whaley approximation can also be used to value European and American cross currency and quanto options with some of modifications of parameters used in pricing formulas.

1. In case of cross currency options, the option price equals P/X_0, where X_0 is the spot FX rate as of the analysis date and P is the price of an ordinary option obtained via the following input transformations:
   - Underlying price: $F \to F$
   - Strike: $K \to X_0 K$,
   - Volatility: $\sigma \to \sqrt{\sigma^2 + \sigma_X^2 - 2\rho\sigma\sigma_X}$,
   - Risk free rate: $r_T \to \tilde{r}_T$,
   - cost-of-carry: $r_T - q \to \tilde{r}_T - q + r_T^{fin,tot} - r_{under\_ccy}^{rf}$
   - Yield parameter remains r_T, but which now yields a non-zero drift (cost of carry).
2. In case of quanto options, the option price equals $P/\bar{X}$, where $\bar{X}$ is the fixed FX rate and P is the price of an ordinary option with the following input transformations:
   - Underlying price: $F \to F$
   - Strike: $K \to K$
   - Volatility: $\sigma \to \sigma$,
   - Risk free rate: $r_T \to \tilde{r}_T$,
   - cost-of-carry: $r_T - q \to r_T^{fin,tot} - q + \rho\sigma\sigma_X$
   - Yield: $r_T \to \tilde{r}_T - \rho\sigma\sigma_X$, yielding a drift (cost of carry) $\rho\sigma\sigma_X$.

For more details about the cross currency and quanto methodologies please see [4].

Because we do not have implied volatility time series for quanto and cross-currency options, we proxy the volatility with the at-the-money volatility of the specified volatility time series (typically the volatility series of the corresponding vanilla option). The volatility is constant across strikes for a given time-to-maturity.

4 Use of Total Return (TR) Equity Time Series

The default behavior of models including equity risk uses one and the same times series of equity levels a) for pricing and security calibration needs, with prices extracted as of pricingDate, and b) for computing equity returns used in risk forecasting or volatility estimates based on historical equity time series. This means that whenever a historical return is needed, for anything like risk, predictive stress test estimates, or calculation of volatility from equity time series for instance, the return from t_1 to t_2 is calculated from the equity level, e.g. in logreturn space $r_{t_1 \to t_2} \equiv \log[S(t_2)/S(t_1)]$.

Since this behavior does not account for the possibility of a dividend payout between the two dates, which could be large enough to bias the computed return[1], a methodology using total return time series is also available in RiskServer/RiskManager. Ifa total dividend payout D is made between t_1 (excluded) and $t_2 > t_1$ (included), the return adjusted for such payout should be

$$R_{t_2}^{TR} = [S(t_2) + D]/S(t_1) - 1, \tag{22}$$

instead of[2]

$$R_{t_2} = S(t_2)/S(t_1) - 1. \tag{23}$$

The methodology using total returns makes use of two distinct time series associated to the same RM equity identifier: one is the traditional time series of equity price levels, which is used whenever the equity price is needed on the pricing date, the other is a time series of adjusted returns which is used for risk forecasting and for the calculation of any analytics involving realized past equity returns (as when a volatility is derived from historical time series of returns, for instance), replacing (23) with the equity total return.

Equity total return time series use the same data used in MSCI Equity Index analytics. While not available for every distinct equity available in Risk Server[3], equity without a TR time series reverts to use the price

[1] In case of exceptional dividends exceeding a certail threshold as a percentage of price, a new equity time series in RiskServer is created and given a distinct identifier, similarly to the case of splits or reverse splits, thus eliminating the potential bias. The same is not done for ordinary dividends, though, which is the reason for introducing the enhanced approach described here.

[2] In log-return space $\log[(S(t_2)+D)/S(t_1)]$ replaces $\log[S(t_2)/S(t_1)]$.

[3] An equity traded on different exchanges, for instance, will use the same return time series, even if closing prices on different exchanges might be slightly different, and might imply slightly different returns if computed directly from price levels. **Also note that not all assets appearing as equities in RiskServer/RiskManager support TR time series. Equity index levels, for instance, correspond to total return or not depending on the specific index, and do not come with an associated TR time series; stock units, as another example, do not yet support this functionality.**

level time series when the TR metholdology is activated. A TR daily equity time series is essentiallly identical, for risk estimation purpose, to the the time series of returns derived from price levels with the exception of a discrete set of days per year associated to dividend payouts, with the difference between the total return and the price return on the ex-dividend date equal to the actual dividend payout expressed as a percentage of the price level before the payout, see eq. (22-23), where t_2 is the ex-dividend date, and D is the dividend.

The methodology using TR time series for risk can be enabled at two different levels: at the general query level by specifying the *enable* choice in equityTotalReturnSettings available in rmlQuery.timeSeriesData; and at the level of the riskSetting specifications where the same equityTotalReturnSettings input can be enabled or disabled in each valuationSpec. If specification is provided at both layers, the riskSetting level has precedence over the general query level.

Total Return time series can be viewed in Risk Manager under Market Data Viewer.

5 Price Calibration

In Risk Manager price calibration is the process by which valuation and risk models are adjusted to match the option price or volatility information provided by the user.

The optional inputs used in the calibration procedure are:

- **MP** - Market price
- **UV** - User specified volatility
- **IVS** - Implied vol time-series identifier

Note that **IVS** is required for implied volatility to be a risk factor, otherwise volatility is assumed to be static, i.e. it is taken to be the same in base and simulated scenarios. Specification of **IVS**, i.e. the volatility time series used for estimation of vega risk, can be explicit via the input *VolatilitySeries*, or it can be obtained via the input *UseHistoricalVolatility* which, if specified, prompts the construction of the volatility time series from historical data. In this case, no interpolation in maturity or delta is necessary when generating new volatility scenarios for the given option.

The possible combinations of inputs, and the corresponding calibration procedures, are listed in Table 1.

MP	UV	IVS	
			1. Volatility computed from historical series of underlying returns, with history and weighting determined by the risk settings. Note that weighting of returns is determined by the decay factor, but volatility is *always* estimated using daily returns, irrespective of the value of the returnHorizon specified in the risk settings. Static volatility risk model.
✓			**2.** Single implied volatility computed by matching the market price. Static volatility risk model. If calibration fails we revert to **1.** above.
	✓		**3.** Volatility used as specified. Static volatility risk model.
		✓	**4.** Volatility is derived via interpolation of the implied vol data from **IVS**, corresponding to analysis date. Interpolation is linear in the time to maturity and quadratic in the delta of the option, see [5] for details of the quadratic interpolation procedure. Volatility risk model is generated from the history of the volatility series, from which a new volatility is derived for each scenario via interpolation.
✓	✓		**5.** User specified volatility inputs is used and market price is disregarded. Static volatility risk model.
✓		✓	**6.** Volatility is derived as in **4.** above. An overall volatility adjustment factor is computed by matching the market price in the base scenario. Volatility risk model is generated from historical data corresponding to the volatility series as in **4.** above, using the adjustment factor derived in the base scenario to adjust the resulting volatility for each scenario. If calibration of the adjustment factor fails we revert to **4.** above.
	✓	✓	**7.** Volatility is derived as in **4.** above. An overall volatility adjustment factor is computed by matching the user volatility specified for the base scenario. Volatility risk model is generated from historical data corresponding to the volatility series as in **4.** above, using the adjustment factor derived in the base scenario to adjust the resulting volatility for each scenario. If calibration of the adjustment factor fails we revert to **4.** above.
✓	✓	✓	**8.** Volatility is derived as in **4.** above and market price is disregarded.

Table 1: Possible combination of inputs determining price calibration

The adjustment factor can be either multiplicative or additive depending on the specific calibration settings for the instrument, see *VolAdjustmentType*.

To match a user-specified market price of a quanto option, we calibrate the underlying volatility as in the non-quanto case. Unlike regular options, the calibration factor applies to both the volatility and to the yield. That is, we replace σ with $\lambda\sigma$ in the Black-Scholes volatility term and in the quanto yield term, where λ

is the multiplicative calibration constant. This dependence of q on σ makes the Black-Scholes quanto price a non-monotonic function of σ. In most cases this is not a problem, because the yield term is of secondary importance after the volatility, but in extreme cases this could lead to difficulty in calibration (e.g. if $\lambda\rho\sigma\sigma_x$ becomes too large).

To match a user-specified market price of a cross-currency option, we calibrate the currency-translated underlying volatility $\sqrt{\sigma^2+\sigma_X^2+2\rho\sigma\sigma_X}$. This means that we apply a single multiplicative constant to both the underlying volatility (σ) and the FX rate volatility(σ_X). Here the underlying volatility refers to the volatility of log-returns of the underlying price *in the underlying currency* (not the payoff currency). The uncalibrated values of σ, σ_X, and ρ are obtained as described in the input table.

Currently the additive volatility adjustment is not available for quanto or cross-currency options.

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